Chapter 4: Role of Institutions

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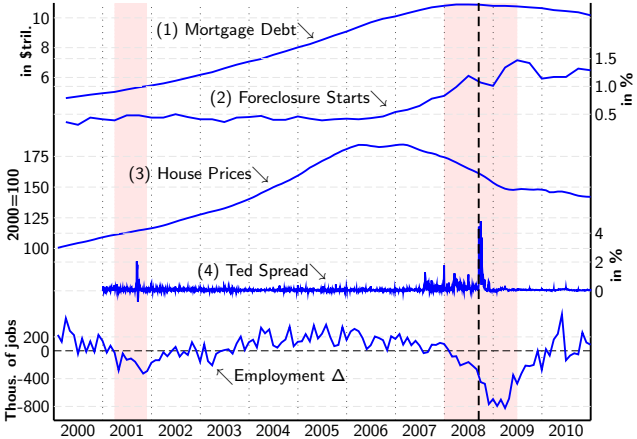
Course Overview



- (1) Debt Growth
- (2) Default
- (3) House Prices
- (4) Financial Crisis
- (5) Institutions
- (6) Expectations

Rating Agencies

- Suppose you are an issuer and you have two securities:

	Peach	**Lemon**
Probability	π	$1-\pi$
Valuation To Issuer	$V_G - \delta$	$V_B - \delta$
Valuation To Investor	V_G	V_B

- Trade always benefits both issuer and investor.
- But Akerlof (1970) showed, only the lemon trades in equilibrium.
- $p = \pi V_G + (1-\pi)V_B$ and only lemon will sell.

Add rating agencies

- Rating agency: credibly identify lemons and peaches.
- Everyone is better off.
  - $V_G - \delta < p_G < V_G$.
  - $V_B - \delta < p_B < V_B$.
- Now suppose that issuer "corrupts" the rating agency.
  - Fee$=\gamma(V_i - V_B)$
  - Rating agency now has incentive to say everything is a peach.
- Is that bad?
  - Yes. Investors realize that rating agency is corrupt.
  - Return to Akerlof (1970) equilibrium.
  - Rating agency earns nothing.
  - Investors and issuers can no longer trade peaches.
- NO ONE BENEFITS! Conflict of interest is BAD FOR EVERYONE.

Bolton, Freixas, and Shapiro (2012)

- Maybe investors didn't realize that rating agencies were corrupt.
- Unlikely

	Institution	Loss ($ billions)		Institution	Loss ($ billions)
1	Citigroup	42.9	11	Washington Mutual	9.1
2	UBS	38.2	12	Credit Agricole	8.3
3	Merrill Lynch	37.1	13	Lehman Brothers	8.2
4	HSBC	19.5	14	Deutsche Bank	7.6
5	IKB Deutsche	15.9	15	Wachovia	7.0
6	Royal Bank of Scotland	15.2	16	HBOS	7.0
7	Bank of America	15.1	17	Bayerische Landesbank	6.7
8	Morgan Stanley	14.1	18	Fortis	6.6
9	JPMorgan Chase	9.8	19	Canadian Imperial (CIBC)	6.5
10	Credit Suisse	9.6	20	Barclays	6.3

Why do we have rating agencies?

- Focus in "Issuer pays" is on asymmetric information between investors and issuers.
- But there is another asymmetric information problem: Fund managers vs. investors:
  - Risk neutral fund manager wants to take on as much risk as possible.
  - Risk averse pension fund wants to limit risk
- Trustees use ratings to identify low risk investments to protect investors.
- Structured Finance (MBS, CDO, etc.) – rating agency had the same information as investors.
- "Investor pays" would actually be "fund manager pays"
  - Another conflict of interest.

Simple Model

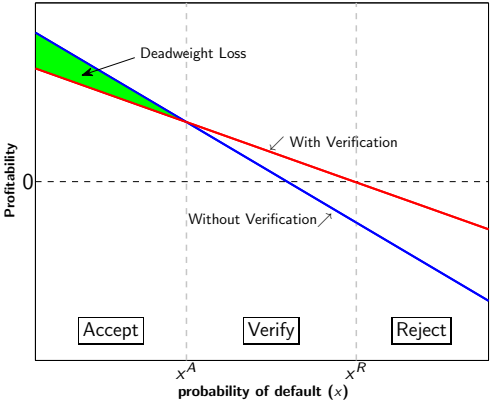
- Suppose lender can observe x
  - index of # of 60DQ in last year.
- For simplicity $x = P(default)$.
- For a pool of N loans, xN will default.
- Suppose by verifying loans, lender can identify s percent of the defaulters. (sxN borrowers identified)
- Assume interest rate is $\overline{R}$ and cost of funds is 1.
- Payoffs

$$\textit{Not Verifying}: \underbrace{(1-x)(\overline{R}-1)}_{\text{Repayers}} + \underbrace{x \times -1}_{\text{Defaulters}} = \overline{R} - 1 - \overline{R}x$$

$$\textit{Verifying}: \underbrace{(1-x)(\overline{R}-1)}_{\text{Repayers}} + \underbrace{sx \times 0}_{\substack{\text{Identified}\\ \text{Defaulters}}} + \underbrace{(1-s)x \times -1}_{\substack{\text{Unidentified}\\ \text{Defaulters}}} - c = \overline{R} - 1 - c - (\overline{R} - s)x$$

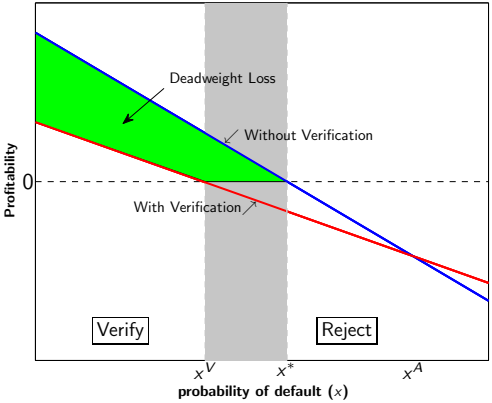
The deadweight loss of ATR

- More defaults = less profits
- Reject loans below x^*
- Introduce verification
  - Reject loans above x^R
  - Accept loans below x^A
  - Verify loans in between
- If we require verification
- Deadweight loss for high credit quality loans
- Cost of verification < benefit.



ATR will block profitable loans

- Suppose we consider a similar scenario
- But suppose verification is much more costly.
  - Self employed borrowers.
- If we force verification
- Only borrowers above x^V get loans
- Large deadweight loss
- Borrowers between x^V and x^A can no longer get loans.
- Documenting income of self-employed is just too expensive.



Justifying ATR

- Purpose of underwriting is to protect the *lender*.
- But it generates a deadweight loss for *lenders*...
- Justification: Protect borrowers.
  - "Behavioral" arguments: Borrowers prefer fewer choices
  - Laibson (1997)
  - Gul and Pesendorfer (2001)
- Or social planner wants fewer foreclosures.
- Return to that later.

Simple Principal Agent Problem

Agent

- expends costly effort a to generate output

$$\tilde{y} = a + \tilde{\eta} \text{ where } \tilde{\eta} \sim \mathrm{N}(0, \sigma^2)$$

- Cost to the agent of effort is

$$c(a) = \frac{1}{2}a^2$$

- Maximizes utility of consumption

$$\begin{aligned} U(\tilde{c}) &= \mathrm{E}\left(-\tfrac{1}{A}\exp -A\tilde{c}\right) \\ &= -\tfrac{1}{A}\exp -A\left(\mathrm{E}(\tilde{c}) - (1/2)A\mathrm{var}(\tilde{c})\right) \end{aligned}$$

Principal

- Wants to maximize output
- Offers incentive scheme:

$$s(\tilde{y}) = \delta + \gamma\tilde{y}$$

- Chooses optimal γ and δ

$$\begin{aligned} \max \pi &= \mathrm{E}(\tilde{y} - s(\tilde{y})) \\ &= (1-\gamma)\mathrm{E}(\tilde{y}) - \delta \end{aligned}$$

Agent's Problem

- Consumption

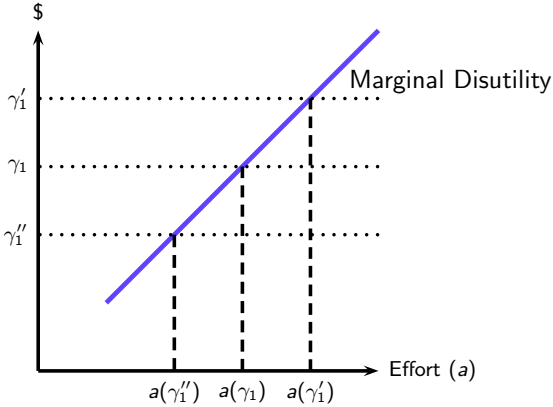
$$\tilde{c} = \gamma\tilde{y} + \delta - \frac{1}{2}a^2$$

- Agent chooses effort

$$\begin{aligned} \max U(\tilde{c}) \quad &\propto \quad \mathrm{E}(\tilde{c}) - (A/2)\mathrm{var}(\tilde{c}) \\ &= \delta + \gamma\mathrm{E}(\tilde{y}) - \tfrac{A}{2}\mathrm{var}(\tilde{c}) - \tfrac{1}{2}a^2 \\ &= \delta + \gamma a - \tfrac{A}{2}\gamma^2\sigma^2 - \tfrac{1}{2}a^2 \end{aligned}$$

- Solution:

$$\partial U/\partial a = 0 \Rightarrow \gamma = a$$



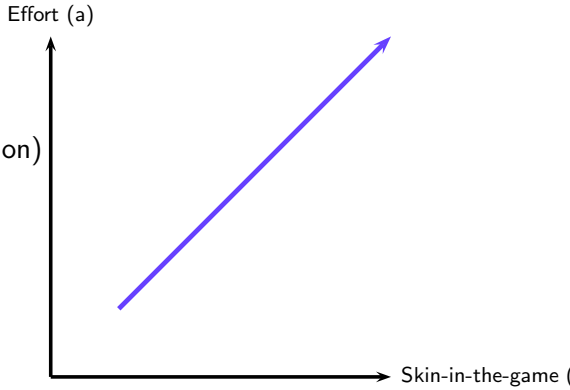
Principal's Problem

- Chooses γ and δ

$$\max(1-\gamma)a - \delta$$
$$\ni \quad a = \gamma \qquad \text{(IC)}$$
$$\delta + \gamma a - \frac{A\sigma^2}{2}\gamma^2 - \frac{1}{2}a^2 = 0 \qquad \text{(Participation)}$$

- Substitute in the constraints:

$$= (1-\gamma)a + \underbrace{\gamma a}_{\text{Incentive}} - \underbrace{\frac{A\sigma^2}{2}\gamma^2}_{\text{Risk}} - \underbrace{\frac{1}{2}a^2}_{\text{Effort}}$$
$$= a - \frac{A\sigma^2}{2}\gamma^2 - \frac{1}{2}a^2$$
$$= \gamma - \frac{A\sigma^2}{2}\gamma^2 - \frac{1}{2}\gamma^2$$



$$\max \quad \underbrace{\gamma}_{\text{benefit of incentives}} - \underbrace{\frac{1}{2}(A\sigma^2+1)\gamma^2}_{\text{cost of incentives}} \Rightarrow \gamma = \frac{1}{A\sigma^2+1}$$

Does the market get this right?

- Equilibrium is not Pareto-optimal
  - Planner sets effort $a = 1$
  - Allocates all risk to principal
- What if planner cannot observe effort?
- Sets incentive γ and δ to maximize:

$$\lambda E(\tilde{y} - \gamma\tilde{y} - \delta) + (1 - \lambda)U(\tilde{c})$$

- Solution: $\gamma = \frac{1}{A\sigma^2+1}$!
- Equilibrium is *constrained* pareto-optimal
- General Proposition: Prescott and Townsend (1984)
  - One good economy
- Multiple goods – relative price effects mean constrained suboptimality
  - Geanakoplos and Polemarchakis (1986)
  - Greenwald and Stiglitz (1986)

Free Market for contracts:

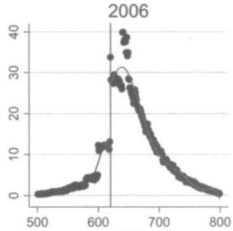
When you spend, you may spend your own money or someone else's; and you may spend for the benefit of yourself or someone else. Combining these two pairs of alternatives gives four possibilities summarized in the following simple table:" (**?**, 116)

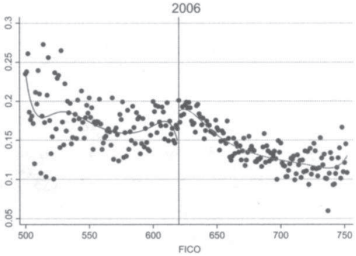
Whose Money	On Whom Spent	
	You	Someone Else
Yours	I	II
Someone Else's	III	IV

"Category IV refers to your spending someone else's money on still another person. You are paying for someone else's lunch out of an expense account. You have little incentive either to economize or to try to get your guest the lunch he will value most highly." (**?**, 117)

Keys et al. (2010)

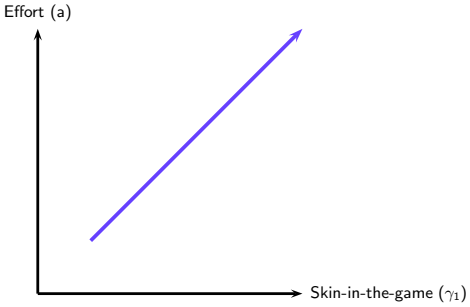
- Argue that there is a "Rule of Thumb"
  - Credit score $< 620 \Rightarrow$ more difficult to securitize
- Comparing 620 and 619, fewer securitized loans and more defaults
  - Lower credit score should mean more default!
- Argue that lenders were more careful because they could not sell loan – more risk retained





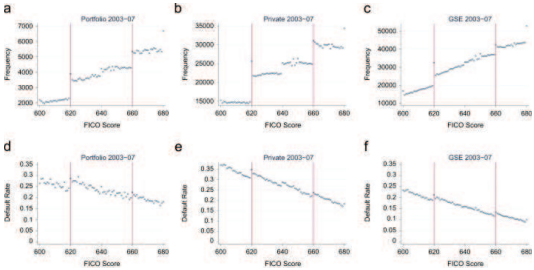
What does this tell us?

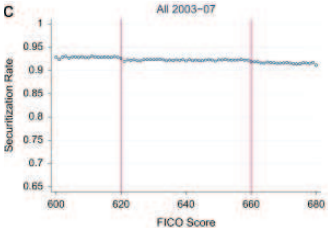
- Higher γ implies more effort
  - Central to the model
- Question is if γ too high or too low
  - No evidence here
- From *House of Debt*
  "Moral hazard refers to a situation in which a sophisticated individual games a flawed system by taking advantage of a naive counterparty. The classic example of moral hazard is someone driving irresponsibly after getting auto insurance because he knows the insurance company will pay for an accident. If the auto insurance company naively provides unlimited insurance at a cheap price, the drivers moral- hazard problem could become quite severe."



Bubb and Kaufman (2014)

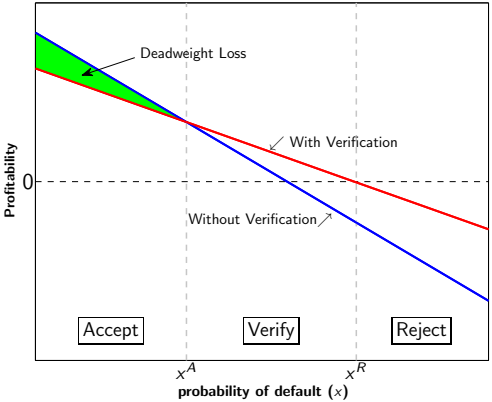
- Using different data, showed that all loans exhibited the same pattern
  - PLS securitized
  - portfolio
  - GSE
- Loans < 620 FICO not more likely to end up on portfolio.





Why do we see cutoff?

- As quality of loans falls, likelihood of verification increases
- At x^A cost of verification = increased cost of default.



The slide you've all been waiting for...

- The end.

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